INT421—Digital Signal Processing

The z-Transformation-II

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Topics of Last Lecture

- The z-transformation
- Convergence region of z-transform.
- Relation between z-transform and difference equation.

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Transfer Function

- The causal discrete-time LTI systems

$$y[n] = -\sum_{l=1}^{N} a_l y[n-l] + \sum_{k=0}^{M} b_k x[n-k]$$

from linearity and translation properties of z-transform

$$Y(z) = -a_1 z^{-1} Y(z) - a_2 z^{-1} Y(z) + \cdots + b_0 X(z) + b_1 z^{-1} X(z) + b_2 z^{-2} X(z) + \ldots$$

- $H(z) = Y(z)/X(z)$ is called the transfer function or **system function** of the system.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \ldots}{1 + a_1 z^{-1} + a_2 z^{-2} + \ldots}$$

- $H(z)$ is the z-transform of $h(n)$.

Quiz

Example

What's the condition for the system (filter) $y[n] = x[n-1] + \alpha y[n-1]$ to be stable?

Solution

$$Z(y[n] = x[n-1] + \alpha y[n-1]) = Y(z)(1 - \alpha z^{-1}) = z^{-1}X(z)$$

The transfer function is:

$$G(z) = \frac{Y(z)}{X(z)} = \frac{1}{z - \alpha}$$

hence for the system (filter) to be stable we need $|z| < 1$.

Quiz

Example

Find impulse response $h[n]$ for a system described by the following input-output relationship: $y[n] = -2y[n-2] + x[n]$:

Solution

Write z-transforms

$$Z(y[n]) = -2z^{-2}Y(z) + X(z), \quad so \quad (1 + z^{-2})Y(z) = X(z) \quad and$$

$$H(z) = \frac{1}{1 - z^{-2}}$$

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Rational Function

- The transfer function of a system represented by difference equation is a rational function.

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2} + \dots}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots} = \frac{\sum_{k=0}^{N} b_k z^{-k}}{1 + \sum_{k=1}^{N} a_k z^{-k}} = \frac{N(z)}{D(z)}$$

- The roots of the denominator $D(z) = 0$ are called **poles** where $X(z) = \infty$.
- The roots of the numerator $N(z) = 0$ are called **zeros** where $X(z) = 0$.
- From the **fundamental theorem of algebra**, we have

$$H(z) = \frac{H_0 \prod_{k=1}^{N} (z - z_k)}{\prod_{k=1}^{N} (z - p_k)}$$

  where the zeros and poles either purely real, or appear in complex conjugate pairs.
- A **pole-zero plot** is a graphic description of rational $X(z)$, up to the scale factor. Use $\circ$ for zeros and $\times$ for poles. Multiple poles or zeros indicated with adjacent number.

Quiz

Example

Consider the Z-transform signal

$$X(z) = \frac{1}{1 - 0.5z^{-1}}$$

Determine the poles and zeros.

Solution

$$X(z) = \frac{1}{1 - 0.5z^{-1}} = \frac{z}{z - 0.5}$$

- *The zeros are:* $\{0\}$
- *The poles are:* $\{0.5\}$

Quiz

Example

Consider the Z-transform signal

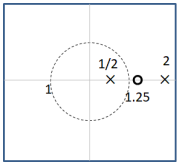
$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 2z^{-1}}$$

Determine the pole–zero plot for the signal.

Solution

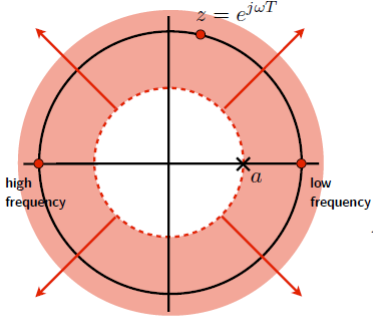
$$X(z) = \frac{z}{z - 0.5} + \frac{z}{z - 2} = \frac{z(z-2) + z(z-0.5)}{(z - 1/2)(z-2)} = \frac{2z(z - 1.25)}{(z - 1/2)(z-2)}$$

- *The zeros are:* $\{0, 1.25\}$ *and the poles are:* $\{0.5, 2\}$



Stability of LTI system

- An LTI system is stable *if and only if* its **poles are inside the unit circle**. In other words, $|p_k| < 1 \quad \forall k = 1, 2, \ldots, N$
- An LTI system is stable iff $H(z)$ exists for $|z| = 1$
  1. **causal system**: if the ROC extends outward from the out most pole
  2. **stable system**: ROC includes the unit circle
  3. **causal and stable system**: all poles must be inside the unit circle



Stability of LTI system

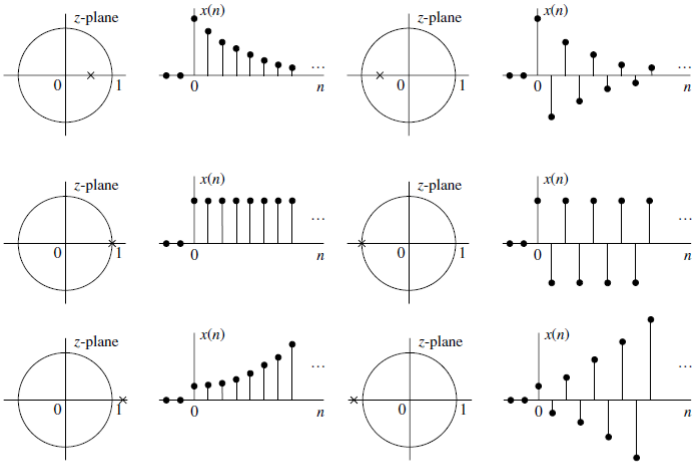


Figure 3.5 Time-domain behavior of a single-real-pole causal signal as a function of the location of the pole with respect to the unit circle.

Quiz

Example

Suppose the transfer function of an LTI system is given by $H(z) = \frac{0.5(z-1)z}{(z-0.5)(z-0.75)}$

1. What are the "zeros" of this system?
2. What are the "poles" of the system?
3. Is this system stable (yes/not)?

Solution

- *The zeros are:* $\{0,\ 1\}$
- *The poles are:* $\{0.5,\ 0.75\}$
- *the system is stable.*

Quiz

Example

Suppose the LTI system is given by $y[n] = 2x[n-1] - 3x[n]$ Check if this system is stable or not.

Solution

Taking z-transform of both sides, get:

$$Y(z) = 2z{-}1X(z) - 3X(z) \quad (\textit{using shift property of } z - \textit{transform})$$

$$H(z) = \frac{(2-3z)}{z}$$

which has a single pole at $z = 0$ (inside the unit circle), hence this system is stable.

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The Inverse of z-Transform

The z-transform $X(z)$ of a given sequence $x[n]$ is unique. Hence the z-transform and its inverse form a transform pair:

$$X(z) = Z(x[n])$$
$$x[n] = Z^{-1}(X(z))$$

There are three different methods obtaining $x[n]$ from a given $X(z)$:

- **Inspection method**: If one has a table of common z-transform pairs, the inverse can be found by *inspection*.
- **Partial fraction expansion**: is restricted to the case where $X(z)$ is in the form of a rational function.
- **Power series expansion**: apply to a larger class of functions.

Inspection Method

Example

If the z-transform is given as a finite-order polynomial, the inverse can be found by inspection.

Solution

$$X(z) = \sum_{n=0}^{N} x[n]z^{-n}$$
$$= x[0] + x[1]z^{-1} + x[2]z^{-2} + \cdots + x[N]z^{-N}$$

$$X(z) = 1 + 2z^{-10} + 3z^{-20} \Rightarrow x[n] = \delta[n] + 2\delta[n-10] + 3\delta[n-20]$$

Inspection Method

Example

If one is familiar with (or has a table of) common z-transform pairs, the inverse can be found by inspection. For example, one can invert the z-transform.

$$X(z) = \left(\frac{1}{1 - 0.5z^{-1}} \right), \quad |z| > 0.5$$

Solution

Using the z-transform pair

$$a^n u[n] \overset{z}{\longleftrightarrow} \frac{1}{1 - az^{-1}}, \quad |z| > |a|.$$

By inspection we recognize that $x[n] = (0.5)^n u[n]$

Partial Fraction Expansion

Example

If the z-transform is given as a finite-order polynomial, the inverse can be found by Partial Fraction Expansion.

Solution

$$X(z) = \frac{1 + z^{-1}}{(1 + 0.5z^{-1})(1 - 0.5z^{-1})}$$

$$X(z) = \frac{1 + z^{-1}}{(1 + 0.5z^{-1})(1 - 0.5z^{-1})}$$

$$= \frac{A}{1 + 0.5z^{-1}} + \frac{B}{1 - 0.5z^{-1}} \Rightarrow X(z) = \frac{-0.5z}{z + 0.5} + \frac{1.5z}{z - 0.5}$$

$$\Downarrow$$

$$x[n] = [-0.5(-0.5)^n + 1.5(0.5)^n]u[n]$$

Partial Fraction Expansion

Determine the partial-fraction expansion of the proper function

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}} \tag{4.16}$$

Solution. First we eliminate the negative powers, by multiplying both numerator and denominator by z^2. Thus

$$X(z) = \frac{z^2}{z^2 - 1.5z + 0.5}$$

The poles of $X(z)$ are $p_1 = 1$ and $p_2 = 0.5$. Consequently, the expansion of the form (4.15) is

$$\frac{X(z)}{z} = \frac{z}{(z-1)(z-0.5)} = \frac{A_1}{z-1} + \frac{A_2}{z-0.5} \tag{4.17}$$

Power Series Method

Example

If the z-transform is given $\frac{N(z)}{D(z)}$, to find the inverse we simply divide the polynomial $D(z)$ by (z) to obtain a possible infinite-order polynomial in negative powers of z^{-1}. The coefficients of this polynomial are the inverse values. Determine the inverse Z-transform of

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}}$$

Solution

By dividing the numerator of X(z) by its denominator, we obtain the power series

$$X(z) = \frac{1}{1 - 1.5z^{-1} + 0.5z^{-2}} = 1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \frac{15}{8}z^{-3} + \frac{31}{16}z^{-4}$$

we conclude that $x[n] = \{1, 3/2, 7/4, 15/8, \dots\}$

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Summary

- Rational Function
- Poles and Zeros of transfer function
- The Inverse of z-Transform

Any Questions?

